Cyclistic Bike-Share Case Study

How Does a Bike-Share Navigate Speedy Success?

Prepared by: Naa Adjeley Mensah-Quaye
Course: Google Data Analytics Professional Certificate
Project: Google Data Analytics Capstone
Date: July 2026

Table of Contents

Table of Figures

1 Introduction

This case study examines how casual riders and annual members use Cyclistic bikes differently. Cyclistic is a fictional bike-share company based in Chicago. The company wants to increase the number of annual memberships because annual members are considered more profitable than casual riders. The analysis will use historical trip data to identify patterns in rider behaviour and provide recommendations that can help Cyclistic convert casual riders into annual members.

2 Ask

2.1 Business Task

The business task is to analyse Cyclistic's historical bike-trip data to identify how casual riders and annual members use the bike-share service differently. The findings will support the development of marketing strategies aimed at converting casual riders into annual members.

2.2 Business Problem

Cyclistic wants to increase the number of annual memberships because annual members are more profitable than casual riders. However, before designing a marketing campaign, the company needs to understand the differences in riding behaviour between the two groups.

2.3 Main Business Question

How do annual members and casual riders use Cyclistic bikes differently?

2.4 Key Stakeholders

Lily Moreno, Director of Marketing
Cyclistic marketing analytics team
Cyclistic executive team
Casual riders and annual members

3 Prepare

3.1 Data Source

The analysis will use publicly available historical bike-trip data provided for the Cyclistic case study. Although Cyclistic is a fictional company, the dataset contains real bike-share trip information and is suitable for comparing the behaviour of casual riders and annual members. The data does not include personally identifiable information, which helps protect rider privacy.

3.2 Data Organisation

The dataset covers 12 months, from July 2025 to June 2026. It consists of 12 monthly CSV files containing individual bike-trip records. The original ZIP and CSV files were stored separately and preserved without modification, while copies were created for cleaning and analysis.

3.3 File Storage Structure

- 01 Original Data
- 02 Cleaned Data
- 03 Analysis
- 04 Visualisations
- 05 Final Report

3.4 Data Credibility and Limitations

The dataset is appropriate for the project because it contains trip-level information that can be used to compare members and casual riders. However, it has some limitations. It does not include personal customer information, reasons for taking trips, membership purchase history, income, age, or place of residence. Therefore, the analysis can identify patterns in bike usage but cannot directly explain why individual riders choose casual passes or annual memberships.

3.5 Relevance to the Business Question

The dataset supports the business question because it includes information such as rider type, trip start and end times, bike type, and station details. These variables can be used to compare ride frequency, ride duration, days of use, and other usage patterns between annual members and casual riders.

4 Process

4.1 Tools Used

Python was used to clean, combine, transform, and analyse the data in Google Colab. Python was selected because the dataset contained nearly six million records, making spreadsheet software less suitable for efficient processing. The pandas library was used to import the files, combine the monthly datasets, inspect data quality, remove invalid records, create new variables, and produce summary statistics.

4.2 Data Cleaning and Transformation

The 12 monthly CSV files covering July 2025 to June 2026 were imported and combined into one dataset containing 5,932,349 rows. A source-file column was added to identify the monthly file from which each record originated.

The started_at and ended_at columns were converted from text into datetime format. No invalid date or time values were identified.

The ride_id column was checked for duplicate values. A total of 35 duplicate ride records were identified and removed, leaving 5,932,314 unique rides.

Missing values were identified mainly in the start and end station names and station identification fields. These records were retained because they still contained valid information about ride duration, rider type, bike type, date, and time. Removing all records with missing station information would have excluded a large portion of the dataset and could have introduced bias into the analysis.

4.3 New Columns Created

The following new variables were created to support the analysis:

- ride_length_minutes: The duration of each ride in minutes.
- day_of_week: The day on which each ride began.
- month: The month in which each ride began.
- hour: The hour of the day when each ride began.
- date: The calendar date on which each ride began.
- source_file: The monthly CSV file from which each record was obtained

4.4 Invalid and Extreme Ride Durations

Ride duration was calculated by subtracting the start time from the end time. The data contained 29 rides with durations of zero minutes or less and 5,532 rides lasting longer than 24 hours. These 5,561 records were removed because they represented invalid or extreme durations that could distort the results. A 24-hour maximum was selected as a reasonable threshold for identifying unusually long trips that could distort the analysis.

4.5 Final Cleaned Dataset

After removing duplicate records and invalid or extreme ride durations, the final cleaned dataset contained 5,926,753 ride records and 19 columns. The cleaned dataset was exported as a CSV file and stored separately from the original data.

5 Analyze

5.1 Rider Distribution

The cleaned dataset contained 5,926,753 valid rides. Annual members accounted for 3,814,650 rides, representing 64.36% of all rides. Casual riders accounted for 2,112,103 rides, representing 35.64% of the total.

This shows that annual members use the service more frequently overall than casual riders.

5.2 Ride Duration

Casual riders took longer trips than annual members. The average ride length for casual riders was 18.57 minutes, compared with 12.07 minutes for members. The median ride length was also higher for casual riders at 11.13 minutes, compared with 8.58 minutes for members.

This suggests that casual riders take fewer trips overall but tend to remain on the bikes for longer periods.

5.3 Usage by Day of the Week

Member rides were strongest during the working week. Tuesday recorded the highest number of member rides, with 618,675 trips. Casual ridership increased sharply during the weekend, with Saturday recording the highest number of casual rides at 445,387 trips.

Casual riders also had their longest average rides during the weekend. Their average ride length was 20.97 minutes on Saturday and 21.59 minutes on Sunday. Member ride lengths were more consistent throughout the week, although they increased slightly during the weekend.

These patterns suggest that members may use the service more regularly for routine travel, while casual riders may be more likely to use the bikes for recreation and leisure. However, this interpretation cannot be confirmed because the dataset does not include the purpose of each trip.

5.4 Monthly Trends

Ridership was highest during the warmer months and declined significantly during winter. Casual ridership peaked in August with 337,246 rides, while member ridership peaked in June with 454,010 rides.

Casual ridership showed a stronger seasonal pattern, with particularly low activity during December and January. Members continued to record considerably more rides than casual users during the colder months, suggesting that member usage is more consistent throughout the year.

5.5 Usage by Time of Day

Both rider groups recorded their highest number of rides at 5:00 p.m. Casual riders recorded 201,168 rides at this hour, while members recorded 411,479 rides.

Member activity also showed a strong morning peak around 8:00 a.m., when 278,840 rides were recorded. The morning and evening peaks among members may indicate regular commuting patterns. Casual usage increased more gradually through the day and was strongest during the afternoon and early evening.

5.6 Bike Type Preference

Electric bikes were the most frequently used bike type among both rider groups. Electric bikes accounted for approximately 70.15% of casual rides and 66.34% of member rides.

Although both groups preferred electric bikes, the preference was slightly stronger among casual riders.

6 Share

6.1 Visualisations

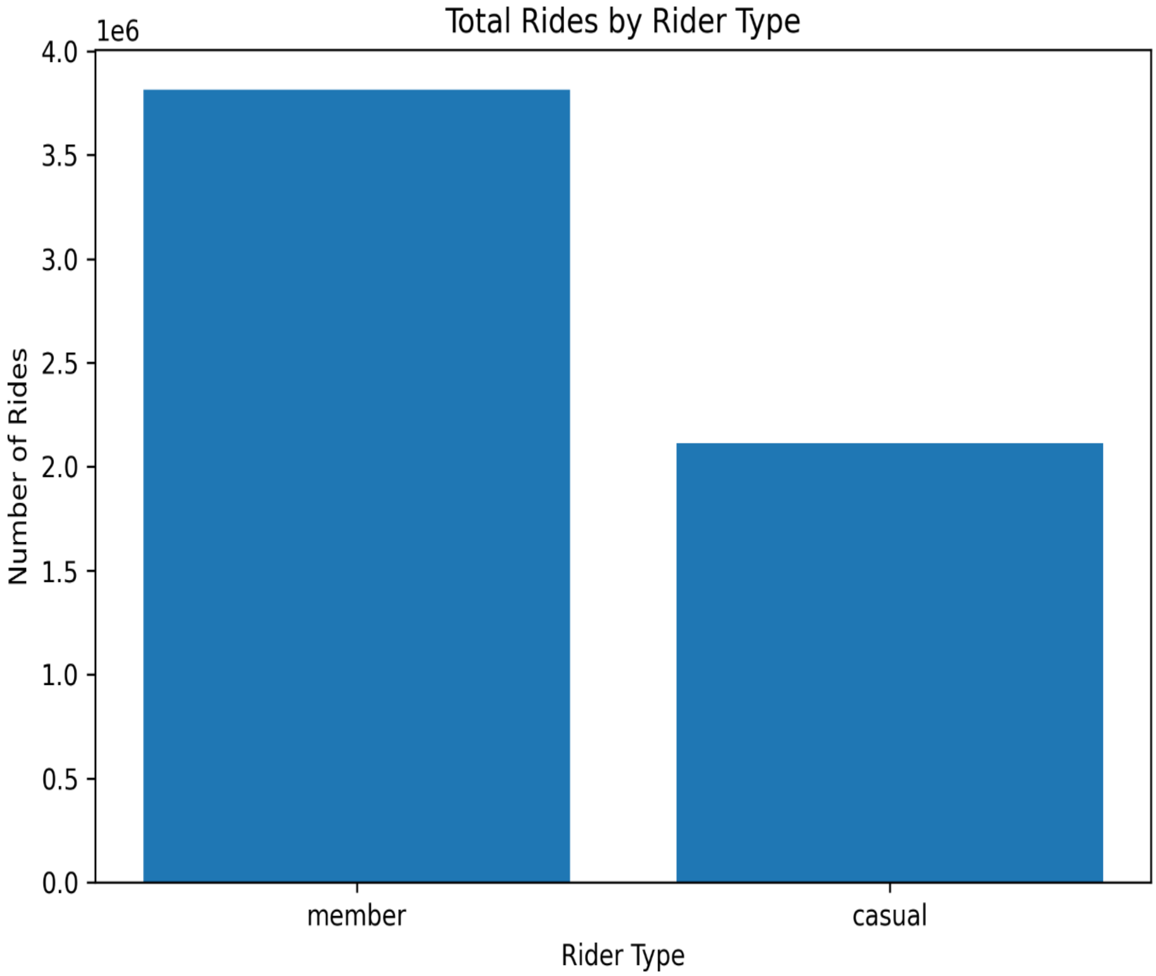

Figure 1: Total Rides by Rider Type

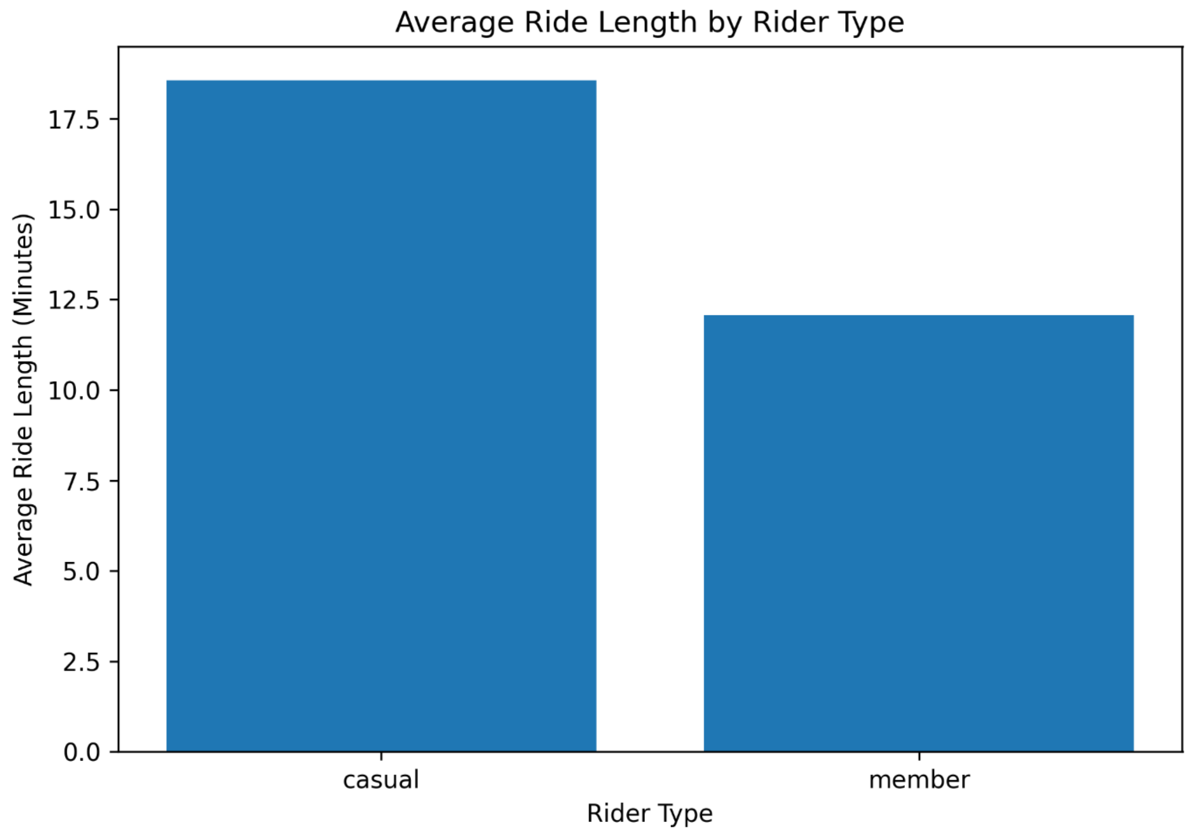

Figure 2: Average Ride Length by Rider Type

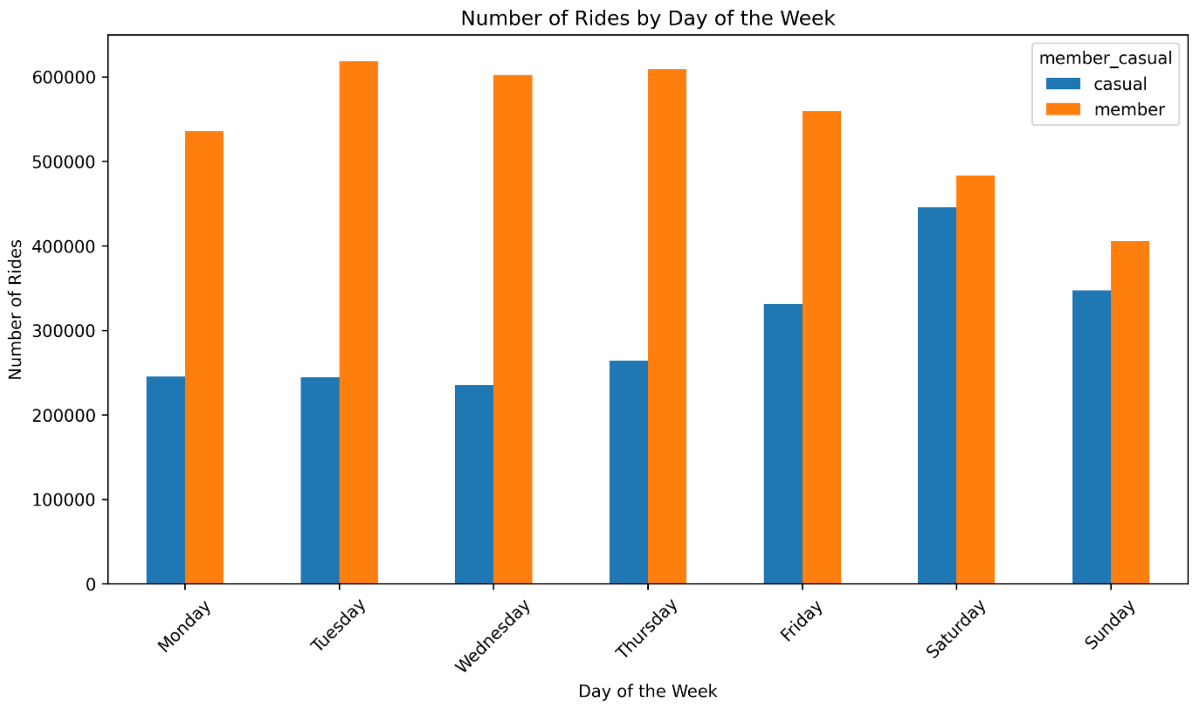

Figure 3: Number of Rides by Day of the Week

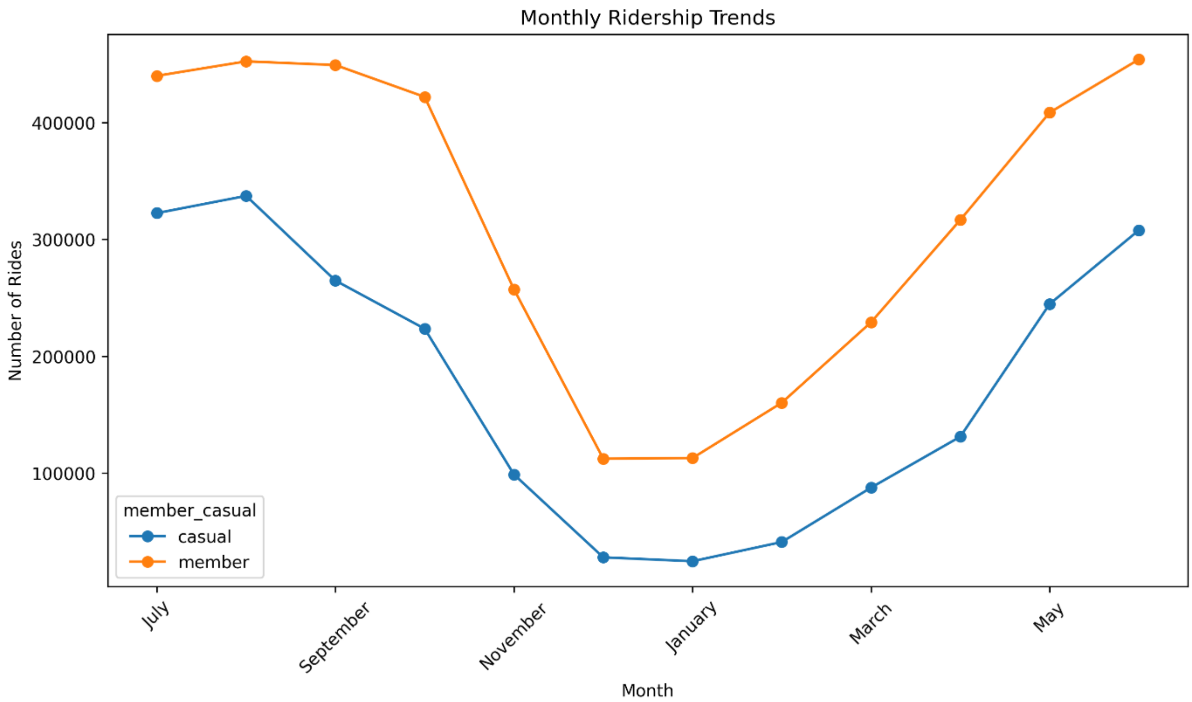

Figure 4: Monthly Ridership Trends

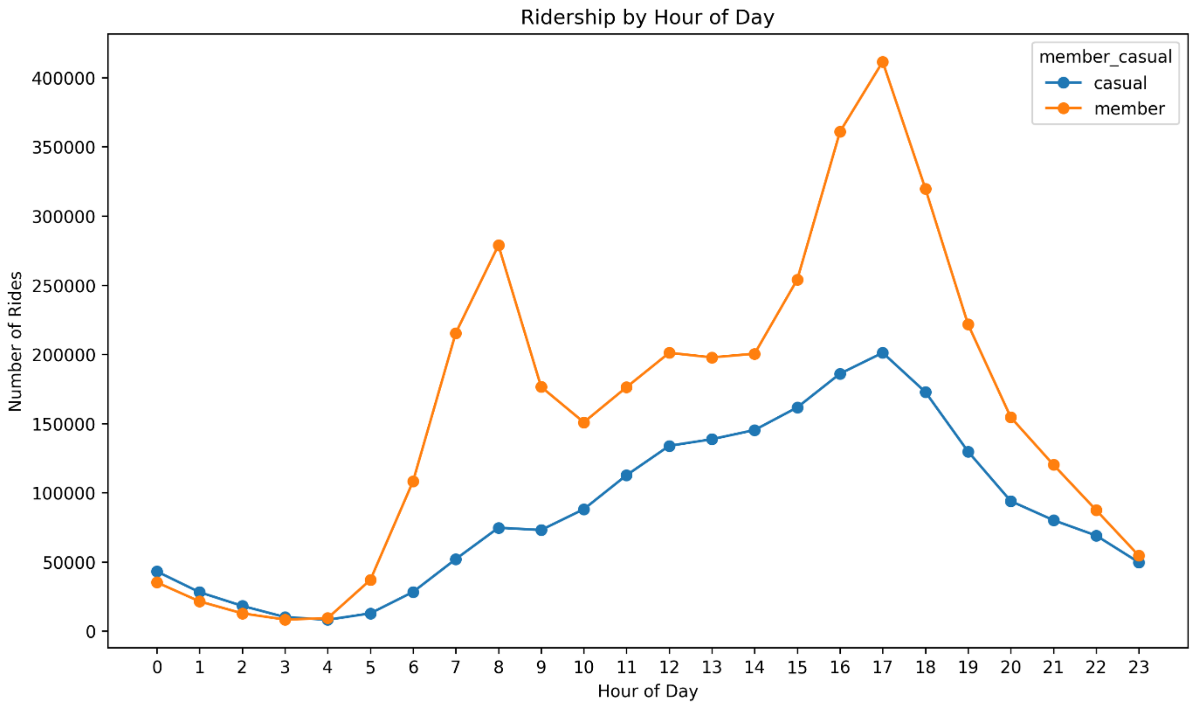

Figure 5: Ridership by Hour of Day

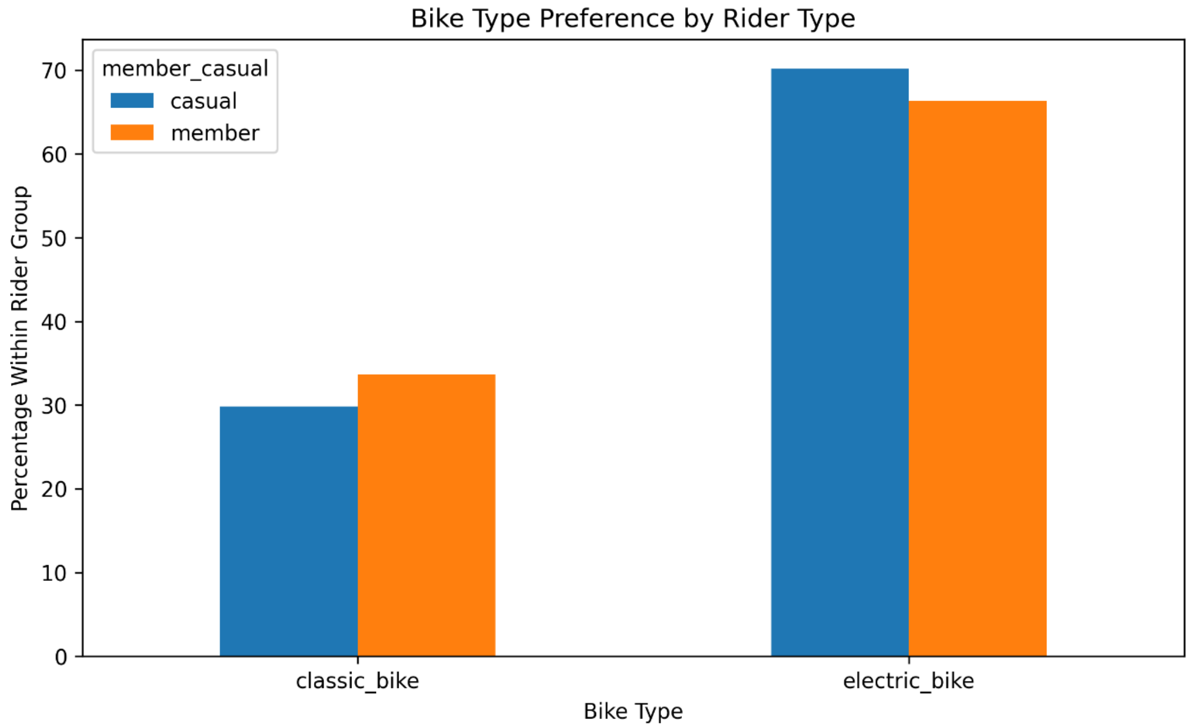

Figure 6: Bike Type Preference by Rider Type

6.2 Key Findings

The visualisations show clear differences between casual riders and annual members. Members account for the majority of rides and use the service more consistently throughout the week and across the year. Their activity is particularly strong during weekday morning and evening periods, which may indicate routine travel or commuting.

Casual riders take fewer rides overall, but their trips are longer on average. Their usage is highest during weekends and warmer months, suggesting that they may be more likely to use the service for leisure or recreational purposes.

Electric bikes are the most commonly used bike type among both groups. The preference is slightly stronger among casual riders, which may provide an opportunity for targeted membership promotions focused on electric-bike users.

6.3 Main Answer to the Business Question

Annual members use Cyclistic bikes more frequently, more consistently, and for shorter trips, particularly during weekdays and peak travel hours. Casual riders use the bikes less frequently but take longer trips, with usage concentrated on weekends, afternoons, and warmer months. These differences suggest that members are more likely to use the service for regular transportation, while casual riders may use it more for leisure.

7 Act

7.1 Conclusion

The analysis found clear differences in how casual riders and annual members use Cyclistic bikes. Members account for most rides and use the service more consistently during weekdays, peak commuting hours, and throughout the year. Casual riders take fewer trips, but their rides are longer and are more concentrated on weekends, afternoons, and warmer months.

These patterns suggest that casual riders may be more interested in leisure and recreational use, while members may rely on the service for more regular transportation. Cyclistic can use these differences to create more targeted campaigns aimed at converting casual riders into annual members.

7.2 Recommendations

Recommendation 1: Promote weekend and seasonal memberships

Cyclistic should target casual riders during weekends and warmer months, when their usage is highest. The company could promote limited-time membership discounts, summer membership offers, or weekend-focused promotions to encourage frequent casual riders to consider annual membership.

Recommendation 2: Target electric-bike users

Electric bikes account for most casual rides. Cyclistic should create digital campaigns that highlight the benefits of annual membership for electric-bike users, such as convenience, frequent access, and potential cost savings for regular riders.

Recommendation 3: Use personalised digital marketing

Cyclistic should use email, mobile notifications, and social media advertisements to reach casual riders during peak usage periods. Marketing messages could focus on how annual membership benefits people who ride regularly on weekends, during the afternoon, or throughout the warmer months.

7.3 Suggested Next Steps

Cyclistic should test the recommended campaigns with smaller groups of casual riders before launching them widely. The company should track measures such as membership conversions, advertisement engagement, repeat usage, and changes in casual-rider behaviour.

Future analysis could also include pricing information, customer surveys, trip purpose, and membership purchase history. These additional data sources would help Cyclistic better understand why casual riders choose not to become members.

8 References

Google Data Analytics Professional Certificate. (2026). *Case Study 1: How Does a Bike-Share Navigate Speedy Success?*

Divvy Bikes. (2025-2026). *Divvy Trip Data: July 2025 to June 2026.*

Motivate International Inc. *Divvy Data Licence Agreement.*